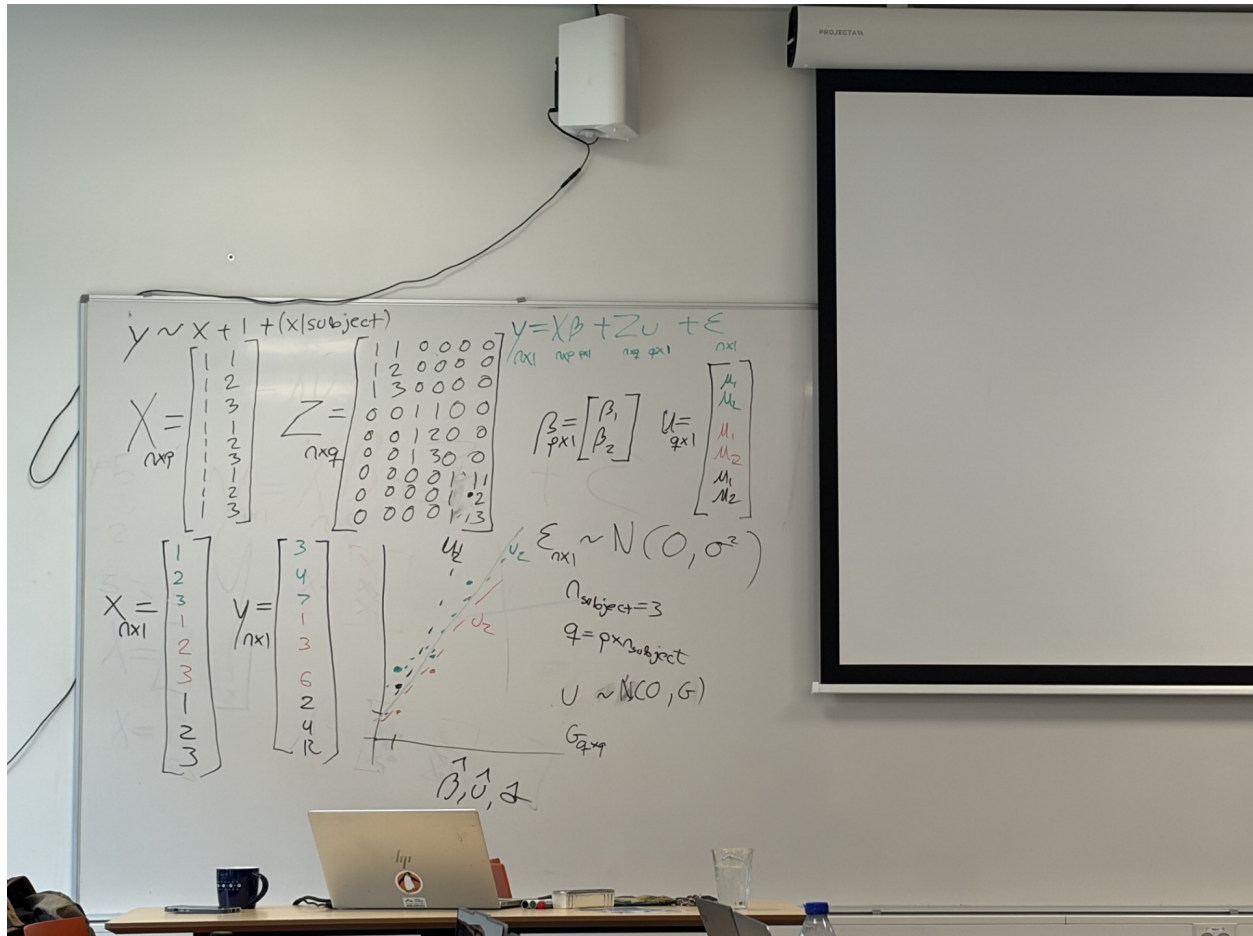


worked_example multilevel

Lau Møller Andersen

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$x_{n \times 1}$ is the independent variable from three different subjects, colour coded.

$y_{n \times 1}$ is the dependent variable, measured from the same subjects, colour coded.

$X_{n \times p}$ is the design matrix for the general effects across participants (first level effects) (intercept and slope)

$Z_{q \times p}$ is the design matrix for the effects for each participant (second level effects) (intercept and slope for each subject)

$\beta_{p \times 1}$ are the coefficients for the two parameters of X

$u_{q \times 1}$ are the coefficients for the six parameters of Z ($q = n_{subjects} \times p$)

$\epsilon_{n \times 1}$ are the residuals, assumed to normally distributed with a mean of 0 and a variance of σ^2

u is normally distributed with a mean of 0 and a variance-covariance structure $G_{q \times q}$